

1 Solving Linear Equations

To solve a linear equation of the form $ax + b = c$, where $a \neq 0$, isolate the linear term ax from the constant term by subtracting b on both sides. Then, divide by a to obtain $x = \frac{c-b}{a}$. For equations with linear and constant terms on both sides, move all linear terms to one side and all constant terms to another, then solve.

Practice Problems:

1. Solve $3x - 5 = 10$.
2. Solve $5x + 2 = 4x - 3$.
3. Solve $7x + 4 = 2x + 14$.
4. Solve $2(x + 3) = 10$.
5. Solve $\frac{x}{2} + 3 = 5$.

Solutions:

1. Adding 5 to both sides gives $3x = 15$, Dividing by 3 on both sides yields $x = 5$.
2. Subtract $4x$ on both sides and 4 from both sides to get $x = -5$.
3. Proceed in a similar fashion to get $x = 2$.
4. Divide out the factor of 2 first and then subtract 3 to obtain $x = 2$.
5. Isolate x by subtracting 3, then multiply by 2. The answer is $x = 4$.

2 Ratios and Proportions

A ratio is a comparison of two quantities (for example, $a : b$). A proportion is an equation stating that two ratios are equal (for example $\frac{a}{b} = \frac{c}{d}$), where b and d are nonzero. You can cross multiply to obtain the relationship $ad = bc$.

Types: Direct, inverse, and joint

Direct: One variable increases, other also increases. The equation can be written in the form $\frac{y}{x} = k$ for some constant k , i.e. $y = kx$.

Inverse: One variable increases, the other decreases. The equation can be written in the form $xy = k$ for some constant k , i.e. $y = \frac{k}{x}$.

Joint: One variable is related to the product of two other variables. The equation can be written in the form $\frac{y}{xz} = k$ for some constant k , i.e. $y = kxz$.

When given a proportions word problem, first write an equation relating the variables in the question. Then find the proportionality constant k first to get the relationship between the variables.

Note: For the practice problems below, the notation $x \propto y$ means that x is proportional to y .

Practice Problems:

1. Solve for x : $\frac{2}{3} = \frac{x}{9}$.
2. If $y \propto x$ and $y = 10$ when $x = 2$, find y when $x = 5$.
3. If x is directly proportional to y and x is 15 when y is 10, find y when x is 25.
4. If $a \propto \frac{1}{b}$ and $a = 4$ when $b = 2$, find a when $b = 5$.
5. The cost of building a house (C) varies jointly with the number of rooms (R) and the square footage (S). If it costs \$300,000 to build a house with 3 rooms and 2000 square feet, how much would it cost to build a house with 4 rooms and 2500 square feet?

Solutions:

1. Simply cross-multiply to get $3x = 18$, thus $x = 6$.
2. $y = kx$ for some constant k . Solving $10 = k \cdot 2$, we get $k = 5$. When x is 5, y is $5(5) = 25$.
3. Equation: $x = ky$. Plugging in the known values and solve for k : $k = \frac{3}{2}$. $x = \frac{3y}{2}$. When $x = 25$, solve for y to get $y = \frac{50}{3}$.
4. Since a is inversely proportional to b , $a = \frac{k}{b}$ for some constant k . We have:

$$4 = \frac{k}{2} \implies 8 = k$$

Therefore, $a = \frac{8}{b}$. When b is 5, $a = \frac{8}{5}$.

5. Equation: $C = kRS$.
Find k : $300,000 = k \cdot 3 \cdot 2000$, so $k = \frac{300,000}{(3 \cdot 2000)} = 50$ New cost: $C = 50 \cdot 4 \cdot 2500 = 500,000$ dollars

3 Quadratics

Key Points:

- Quadratic equation (standard form): $ax^2 + bx + c = 0$.
- Vertex form: $f(x) = a(x - h)^2 + k$, where (h, k) represents the vertex coordinates

- **Axis of symmetry:** A vertical line through the vertex, with the equation $x = \frac{-b}{2a}$
- **Vertex:** The turning point of the parabola, either the minimum point (if parabola opens upwards) or maximum point (if parabola opens downwards)
- **Finding the vertex:** The x-coordinate of the vertex is $\frac{-b}{2a}$.
- **Solutions using the quadratic formula:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Detailed Explanation: To solve a quadratic equation, you can use one of the following methods:

- **Factoring:** Rewrite the quadratic as a product of two binomials. For example, $x^2 - 5x + 6 = (x - 2)(x - 3)$.
- **Quadratic Formula:** The solutions to $ax^2 + bx + c = 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- **Completing the Square:** Rewrite the quadratic into the form $(x - p)^2 = q$, then solve for x .

The discriminant $\Delta = b^2 - 4ac$ determines the nature of the roots:

- $\Delta > 0$: Two distinct real roots.
- $\Delta = 0$: One real root (repeated).
- $\Delta < 0$: Two complex roots.

Practice Problems:

1. Solve $x^2 - 4x + 4 = 0$.
2. Solve $2x^2 - 3x - 2 = 0$.
3. Find the vertex of $y = x^2 - 6x + 9$.
4. Factorize $x^2 - 5x + 6$.
5. Determine the discriminant of $x^2 + 4x + 5$.

Solutions:

1. $x^2 - 4x + 4 = 0$: This can be factored as $(x - 2)^2 = 0$, so $x = 2$ is the repeated root.

2. $2x^2 - 3x - 2 = 0$: Using the quadratic formula:

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(-2)}}{2(2)} = \frac{3 \pm \sqrt{25}}{4},$$

$$x = 2, x = -\frac{1}{2}.$$

3. Vertex of $y = x^2 - 6x + 9$: Rewriting as $(x - 3)^2$, the vertex is $(3, 0)$.
4. Factorize $x^2 - 5x + 6$: $x^2 - 5x + 6 = (x - 2)(x - 3)$.
5. Discriminant of $x^2 + 4x + 5$: $\Delta = 4^2 - 4(1)(5) = 16 - 20 = -4$, so there are two complex roots.

4 Sequences and Series

Basic Explanation: A sequence is an ordered list of numbers, and a series is the sum of terms of a sequence. Common types of sequences include arithmetic and geometric sequences.

- **Arithmetic Sequence:** Each term is obtained by adding a constant difference d . The n th term is given by:

$$a_n = a_1 + (n - 1)d.$$

The sum of the first n terms is:

$$S_n = \frac{n}{2}(a_1 + a_n).$$

- **Geometric Sequence:** Each term is obtained by multiplying the previous term by a constant ratio r . The n th term is given by:

$$a_n = a_1 r^{n-1}.$$

The sum of the first n terms is:

$$S_n = a_1 \frac{1 - r^n}{1 - r}, \quad r \neq 1.$$

For an infinite geometric sequence, the sum is:

$$S = \frac{a_1}{1 - r}$$

provided that $|r| < 1$.

Practice Problems:

1. Find the 10th term of the sequence $2, 5, 8, \dots$

- Find the sum of the first 20 terms of the sequence $3, 6, 9, \dots$.
- Find the 5th term of the geometric sequence $1, 2, 4, \dots$.
- Find the sum of the first 10 terms of the geometric sequence $1, 3, 9, \dots$.
- Determine if 150 is a term in the sequence $7, 14, 21, \dots$.
- Find the sum of the following: $1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \dots$.

Solutions:

- $a_{10} = 2 + (10 - 1)3 = 29$.
- $S_{20} = \frac{20}{2}(3 + 60) = 630$.
- $a_5 = 1 \cdot 2^4 = 16$.
- $S_{10} = 1 \frac{1-3^{10}}{1-3} = 29524$.
- $150 = 7 + (n - 1)7$: Solving gives $n = 22$, so 150 is a term.
- $S = \frac{1}{1-\frac{2}{3}} = 3$.

5 Exponents and Logarithms

Exponents and logarithms are inverse operations. Exponents involve repeated multiplication, while logarithms solve for the exponent in expressions like $a^x = b$.

- **Exponents:** Basic rules include:

$$\begin{aligned} a^m \cdot a^n &= a^{m+n}, \\ \frac{a^m}{a^n} &= a^{m-n}, \\ (a^m)^n &= a^{mn}. \end{aligned}$$

- **Logarithms:** Basic rules include:

$$\begin{aligned} \log(ab) &= \log a + \log b, \\ \log\left(\frac{a}{b}\right) &= \log a - \log b, \\ \log(a^n) &= n \log a. \end{aligned}$$

Logarithms solve $a^x = b$ as $\log_a b = x$.

Practice Problems:

- Simplify $2^3 \cdot 2^5$.
- Solve for x : $3^x = 81$.

3. Simplify $\log(10^5)$.
4. Solve for x : $\log_2 x = 5$.
5. Simplify $4^{\log_2(5)}$.

Solutions:

1. $2^3 \cdot 2^5 = 2^{3+5} = 2^8 = 256$.
2. $3^x = 81 \implies x = 4$.
3. $\log(10^5) = 5 \log 10 = 5$.
4. $\log_2 x = 5 \implies x = 2^5 = 32$.
5. $4^{\log_2(5)} = (2^2)^{\log_2(5)} = 2^{2 \cdot \log_2(5)} = (2^{\log_2(5)})^2 = 5^2 = 25$.